

## ORIGINAL ARTICLE

## Pediatric Obesity

# Concurrent and prospective associations of social media usage with binge eating symptoms in early adolescence

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## Abstract

**Objective:** This study investigated longitudinal relationships between social media usage and binge eating (BE) in early adolescence and explored potential moderating effects of sex and BMI.

**Methods:** Data from the Adolescent Brain Cognitive Development (ABCD) Study, including 1940 participants aged 11 to 12 years at Wave 1, were analyzed over three annual waves. Social media addiction (SMA) scores and time spent on social media were calculated from self-report questionnaires. BE symptoms were evaluated using the Kiddie Schedule for Affective Disorders and Schizophrenia (KSADS-5) through self-reports and parent reports. Modified Poisson regression was used to estimate associations, adjusting for demographic and behavioral covariates.

**Results:** SMA symptoms, but not time on social media, were cross-sectionally and prospectively associated with BE. Prospective associations were stronger in male adolescents and differed by baseline body weight. The consistently high SMA group showed the strongest association with follow-up BE compared with the consistently low SMA group, whereas decreased and increased SMA groups showed similarly positive associations.

**Conclusions:** SMA scores predict BE in early adolescence, with differentiation in the relationship by sex and weight status. Further observational and interventional research could illuminate underlying mechanisms and test the value of targeting social media-addictive behaviors to mitigate the risk of disordered eating.

## INTRODUCTION

Binge eating (BE) is characterized by recurrent consumption of large quantities of food in a short time period, accompanied by a sense of loss of control over eating. Obesity and excessive weight are estimated to be three to six times more prevalent in individuals with BE compared with those without [1]. BE is a core feature of BE disorder (BED), which is one of the most prevalent psychiatric conditions among youth and is associated with psychological distress, reduced quality of life, and

comorbidities including obesity, type 2 diabetes, and cardiovascular disease [2].

Social media use is common among youth, with international data demonstrating a 7.4% prevalence of problematic social media use in adolescents [3] and an association with adverse mental health outcomes [4]. Social media usage (defined here as the manner in which social media is used, i.e., including time spent on social media but also other aspects of social media use) has been shown to have direct and indirect effects on BE frequency in studies that have measured the content or time spent on social media (SMT) [5]. For example,

adolescents at greater risk of disordered eating behaviors engage more heavily with highly visual social media content [6], which can increase BE by worsening body dissatisfaction [7]. Previous cross-sectional research in adolescents found that Tumblr use was positively associated with BE [8] and that increased investment in photos and engagement with others' selfies were associated with BED [9]. However, studies of longitudinal relationships between social media usage and BE in adolescents, especially early adolescence, a crucial developmental period, have been scarce. Only one longitudinal study found that increased SMT was associated with higher BED prevalence at a 1-year follow-up in 9- to 10-year-old individuals [10].

Because social media use can have diverse effects, it is important to distinguish between the effects of harmful and nonharmful usage on BE. Social media addiction (SMA) is not currently recognized as a diagnosis in the *Diagnostic and Statistical Manual of Mental Disorders* (Fifth Edition, Text Revision) (DSM-5-TR), but social media-addictive behaviors, i.e., those characterized by compulsive and uncontrollable use of social media [11], may have more profound psychological impacts than SMT due to their potential to disrupt emotional regulation and increase exposure to harmful content [12]. Alternatively and additionally, measures of SMA may show closer associations with BE due to a shared underlying biobehavioral vulnerability that manifests in a co-occurrence of both behaviors. For example, although BED is not classified as an addiction in the DSM-5-TR, BE shares notable similarities with addictive behaviors phenomenologically and mechanistically [13, 14] and sometimes co-occurs with other behavioral addictions such as gambling disorder [15]. Characterizing the distinct associations of SMA measures versus regular social media usage with BE within a longitudinal study design is therefore warranted.

Moreover, the possibility of variation in the impact of social media usage on BE by sex and body mass index (BMI) requires exploration. Because male adolescents experience higher impulsivity and emotional dysregulation due to social media use [16], male adolescents with high social media use might be more susceptible to BE. Alternatively, because the prevalence of BE in early adolescence is comparable between male and female adolescents [17], contrasting with the overall higher prevalence of BE in female adolescents [18], there may be no sex differences in effects. Individuals with higher BMI may experience greater body dissatisfaction and social comparison, which could be aggravated by social media exposure, making them more vulnerable to BE [19]. The role of BMI in moderating the relationships between BE and SMA, a more intense form of engagement with social media, remains unclear.

Using data from a large US national adolescent cohort, we therefore aimed to investigate longitudinal relationships of different measures of social media usage (i.e., SMA and SMT) with BE in early adolescence, as well as the potential moderating effects of sex and BMI on these relationships. Our primary hypotheses were that SMA scores would be prospectively and cross-sectionally associated with BE in early adolescence, and that those showing consistently high SMA would show the highest BE. Additionally, we expected more pronounced associations in male adolescents

## Study Importance

### What is already known?

- Social media use has been associated with disordered eating behavior.

### What does this study add?

- Social media addiction (SMA) scores showed cross-sectional and prospective associations with binge eating (BE).
- Prospective associations of SMA scores with BE were stronger in male adolescents and differed by baseline body weight.
- A history of SMA symptoms was related to follow-up BE, regardless of current SMA status.

### How might these results change the direction of research or the focus of clinical practice?

- Targeted interventions to decrease social media-addictive behaviors could be explored for effects on BE.
- Comprehensive approaches to identify and address co-occurring social media-addictive behaviors and BE in early adolescence may benefit mental health.

and those with higher baseline BMI values. Our overarching goals were to deepen understanding of relationships between social media usage and BE and potentially inform development of targeted interventions to mitigate risk of BE during this critical developmental period.

## METHODS

### Participants

Data for this study came from the Adolescent Brain Cognitive Development (ABCD) Study, which recruited a diverse sample of children aged 9 to 10 years from 21 US sites, closely approximating national demographics for age, sex, race and ethnicity, and socioeconomic status [20]. Baseline demographics and data from the 2-year, 3-year, and 4-year follow-ups (henceforth referred to as Waves 1–3) from the ABCD 5.1 data release (collected from September 1, 2016, to January 15, 2022) were analyzed in the current study. Among 4754 participants included in the ABCD Study 4-year follow-up, we excluded those with missing data on exposure, outcomes, and covariates (with the exception of household income and BMI), resulting in a total sample of 1940 (Figure S1). The research protocol received approval from institutional review

boards at each study site, with central approval by the University of California, San Diego. All child participants provided verbal assent, whereas parents gave written informed consent.

## Measures

### Social media usage

From Waves 1 to 3, youth were surveyed about their social media use on weekdays and weekends. The average daily SMT was calculated using the following formula:  $([\text{weekday time} \times 5] + [\text{weekend time} \times 2])/7$ . Additionally, youth were asked about their number of social media accounts. Participants with  $\geq 1$  account were asked to complete a six-item SMA questionnaire (Cronbach  $\alpha = 0.84$ ), adapted from the Bergen Facebook Addiction Scale [21] to assess symptoms of problematic use. Example items include “I feel the need to use social media sites/apps more and more” and “I use social media sites/apps so much that it has had a bad effect on my schoolwork or job.” Responses were given on a Likert scale from 1 (never) to 6 (very often). Based on their total scores, participants were categorized into the following three groups: no symptoms (score of 6); mild symptoms (scores 7–23); and addiction (score of  $\geq 24$ , indicating problematic social media use) [22].

### BE symptoms

Participants' BE symptoms were assessed with both self-reports and parent reports at Waves 1 and 3 using the Kiddie Schedule for Affective Disorders and Schizophrenia (KSADS-5) [23]. Presence of BE symptoms was defined as any episode of losing control and consuming more than needed while eating in the past 2 weeks. Frequent BE was defined as either the child or parent reporting BE occurring at least once per week for 3 months and was coded as yes or no. Given that reports from youth and parents are largely discordant [24, 25], we generated three types of outcome variables, i.e., combined report (case is identified if reported by either the youth or the parent), child report, and parent report, as recommended in a previous study, to give a comprehensive overview of BE rates in the sample [26]. In order to reduce the likelihood of underestimating symptoms, the combined report was used for the main analysis.

### Covariates

Sex (female or male), race and ethnicity (American Indian, Asian, Black, Hispanic or Latinx, White, or other), household income (<\$50,000, \$50,000–\$100,000, \$100,000–\$200,000, >\$200,000, or not available), and highest parental educational level (below undergraduate, undergraduate, or above undergraduate) were collected at baseline. BMI was calculated from measured weight and height, and the R package *cdcanthro* was used to calculate Center for Disease Control

and Prevention (CDC) age- and sex-specific BMI percentiles. Individuals were categorized as having underweight/healthy weight (<85th percentile) or overweight/obesity ( $\geq 85$ th percentile). Sleep duration was reported by parents using the Parent Sleep Disturbance Scale for Children, and children with 9 to 11 h of sleep were considered to meet sleep duration recommendations. Physical activity was self-reported using the Youth Risk Behavior Survey, with children who were active for at least 60 min 7 days/week considered to meet physical activity recommendations. Screen time was calculated similarly to SMT, with participants who spent  $\leq 2$  h daily on screens considered to meet screen time recommendations. The aforementioned covariates are reported due to previously described associations with either social media use or BE.

## Statistical analysis

ANOVA and  $\chi^2$  tests compared demographic and behavioral covariates between BE and non-BE participants at Waves 1 and 3, as well as demographics and BE status of the study sample compared with all ABCD Study participants and the sample before excluding those not reporting having a social media account. Percentages or means and standard deviations (SD) were calculated for demographics within all participants and the BE and non-BE groups. In order to visualize SMT and SMA trends over the three waves by Wave 1 and 3 BE status, participants were categorized into four BE profiles based on the binary presence of BE symptoms at the two waves. Modified Poisson regression with robust error variance estimated associations of SMT and SMA with BE. For the cross-sectional associations at Wave 1 and 3, prevalence ratios (PR) with 95% confidence intervals (CI) were calculated. For the prospective association between Wave 1 SMT/SMA and Wave 3 BE, relative risks (RR) with 95% CI were calculated. When a prospective association between social media usage (SMT/SMA) and BE was apparent, latent class trajectory modeling was conducted using the *lcmm* package in R to identify distinct trajectory groups in social media usage over the three waves. An initial model with a quadratic term was specified to capture nonlinear trends. Models with one to five latent classes were subsequently fitted, optimizing model parameters to ensure robust starting values and convergence. The optimal number of classes was determined based on the Bayesian information criterion (BIC; lower values indicate better fit), entropy (higher values indicate better classification), and integrated completed likelihood (ICL; lower values indicate better fit, with ICL1 balancing model fit and classification uncertainty and ICL2 further penalizing classification uncertainty), along with clinical interpretability [27]. Modified Poisson regression was then used to estimate the RR of the associations between SMA trajectory groups and BE symptoms at Wave 3, as well as the frequency of BE at Wave 3 among those reporting BE.

In addition to the crude model, stepwise adjusted results are presented to increase the specificity of our findings by progressively controlling for potential confounders. Model 1 adjusted for demographic covariates, Model 2 additionally adjusted for BMI percentiles and

behavioral covariates, and Model 3, used for the longitudinal analysis, additionally adjusted for Wave 1 BE.

In order to explore potential moderators of a prospective association between social media variables and BE, we tested interactions of sex and BMI percentiles with SMT/SMA on BE. In order to aid understanding

and interpretation of findings, we further conducted stratified analyses by sex and BMI group. For sensitivity analyses, we investigated associations of SMT/SMA with BE based on child and parent reports separately to explore potential reporter bias. Owing to the fact that our main analysis of SMA was restricted to participants reporting having a social media

**TABLE 1** Demographic and behavioral characteristics of the participants by BE status at Wave 1.

|                                                            | All participants<br><i>n</i> = 1940 | No BE<br><i>n</i> = 1691 | With BE<br><i>n</i> = 249 | <i>p</i> value |
|------------------------------------------------------------|-------------------------------------|--------------------------|---------------------------|----------------|
| Age, mean (SD)                                             | 11.6 (0.50)                         | 11.6 (0.50)              | 11.6 (0.48)               | 0.161          |
| Sex, <i>n</i> (%)                                          |                                     |                          |                           | 1              |
| Male                                                       | 875 (45.1)                          | 763 (45.1)               | 112 (45.0)                |                |
| Female                                                     | 1065 (54.9)                         | 928 (54.9)               | 137 (55.0)                |                |
| Race/ethnicity, <i>n</i> (%)                               |                                     |                          |                           | 0.131          |
| White                                                      | 961 (49.5)                          | 849 (50.2)               | 112 (45.0)                |                |
| Black                                                      | 267 (13.8)                          | 239 (14.1)               | 28 (11.2)                 |                |
| Hispanic                                                   | 468 (24.1)                          | 393 (23.2)               | 75 (30.1)                 |                |
| Asian                                                      | 39 (2.01)                           | 34 (2.01)                | 5 (2.01)                  |                |
| Other                                                      | 205 (10.6)                          | 176 (10.4)               | 29 (11.6)                 |                |
| Annual household income, <i>n</i> (%)                      |                                     |                          |                           | 0.108          |
| <\$50,000                                                  | 592 (30.5)                          | 518 (30.6)               | 74 (29.7)                 |                |
| \$50,000–\$100,000                                         | 516 (26.6)                          | 440 (26.0)               | 76 (30.5)                 |                |
| \$100,000–\$200,000                                        | 500 (25.8)                          | 443 (26.2)               | 57 (22.9)                 |                |
| >\$200,000                                                 | 195 (10.1)                          | 177 (10.5)               | 18 (7.23)                 |                |
| Not available                                              | 137 (7.06)                          | 113 (6.68)               | 24 (9.64)                 |                |
| Parental education (highest), <i>n</i> (%)                 |                                     |                          |                           | 0.965          |
| Below undergraduate                                        | 851 (43.9)                          | 743 (43.9)               | 108 (43.4)                |                |
| Undergraduate                                              | 524 (27.0)                          | 455 (26.9)               | 69 (27.7)                 |                |
| Graduate                                                   | 565 (29.1)                          | 493 (29.2)               | 72 (28.9)                 |                |
| BMI percentile, mean (SD)                                  |                                     | 66.9 (28.4)              | 80.3 (23.3)               | <0.001         |
| BMI category, <i>n</i> (%)                                 |                                     |                          |                           | <0.001         |
| Underweight/healthy weight                                 | 1132 (58.4)                         | 1037 (61.3)              | 95 (38.2)                 |                |
| Overweight/obesity                                         | 783 (40.4)                          | 634 (37.5)               | 149 (59.8)                |                |
| Not available                                              | 25 (1.29)                           | 20 (1.18)                | 5 (2.01)                  |                |
| Meeting recommended sleep duration, <i>n</i> (%)           |                                     |                          |                           | 0.736          |
| No                                                         | 1052 (54.2)                         | 914 (54.1)               | 138 (55.4)                |                |
| Yes                                                        | 888 (45.8)                          | 777 (45.9)               | 111 (44.6)                |                |
| Meeting recommended physical activity levels, <i>n</i> (%) |                                     |                          |                           | 0.968          |
| No                                                         | 1665 (85.8)                         | 1452 (85.9)              | 213 (85.5)                |                |
| Yes                                                        | 275 (14.2)                          | 239 (14.1)               | 36 (14.5)                 |                |
| Meeting recommended screen time limits, <i>n</i> (%)       |                                     |                          |                           | 0.445          |
| No                                                         | 1607 (82.8)                         | 1396 (82.6)              | 211 (84.7)                |                |
| Yes                                                        | 333 (17.2)                          | 295 (17.4)               | 38 (15.3)                 |                |

**Note:** Presence of BE symptoms was defined as any episode of losing control and consuming more than needed while eating in the past 2 weeks, as reported by either the child or parent. *P* values were calculated for comparisons of participants without and with BE. Sleep duration was reported by parents using the Parent Sleep Disturbance Scale for Children, asking, “How many hours of sleep does your child get on most nights?” Children who received 9 to 11 h of sleep were considered to meet the sleep duration recommendations. Physical activity data were collected from the self-report Youth Risk Behavior Survey, with participants who were active for at least 60 min 7 days/week defined as meeting the physical activity recommendations. Screen time was calculated similarly to time spent on social media, with participants who spent ≤2 h daily on screens defined as meeting the screen time recommendations.

Abbreviation: BE, binge eating.

account, we also repeated analyses of associations between SMT and BE in a larger sample that included participants regardless of whether they reported having a social media account.

We used R (version 4.3.1) for statistical analyses with a two-sided  $p < 0.05$  set for statistical significance.

## RESULTS

### Participants' sociodemographic and behavioral characteristics

Compared with ABCD Study baseline data, participants in this study included a higher proportion of female adolescents, as well as a higher proportion of adolescents with lower socioeconomic status (Table S1). Mean (SD) age of participants at Wave 1 was 11.6 (0.50) years. The sex distribution for BE status was comparable at Wave 1, but, at Wave 3, significantly more female adolescents reported BE. The number of male adolescents reporting BE decreased from 112 (45.0%) at Wave 1 to 80 (33.8%) at Wave 3, whereas the number of female adolescents reporting BE increased from 137 (55.0%) to 157 (66.2%).

Participants with BE were more likely to be categorized as having overweight/obesity at Wave 1, with ~60% falling into this category at both Wave 1 and 3. Both BE and non-BE participants similarly adhered to sleep duration and physical activity recommendations. However, many Wave 3 BE participants exceeded screen time limits at Wave 1, with only 25 (10.5%) adhering to the 2-h daily limit compared with 308 (18.1%) of non-BE participants (Table 1; Table S2).

### Social media usage and BE

Figure 1 shows trends in social media usage across the following four BE profiles: no BE at both waves ( $n = 1538$ ); developed BE ( $n = 153$ );

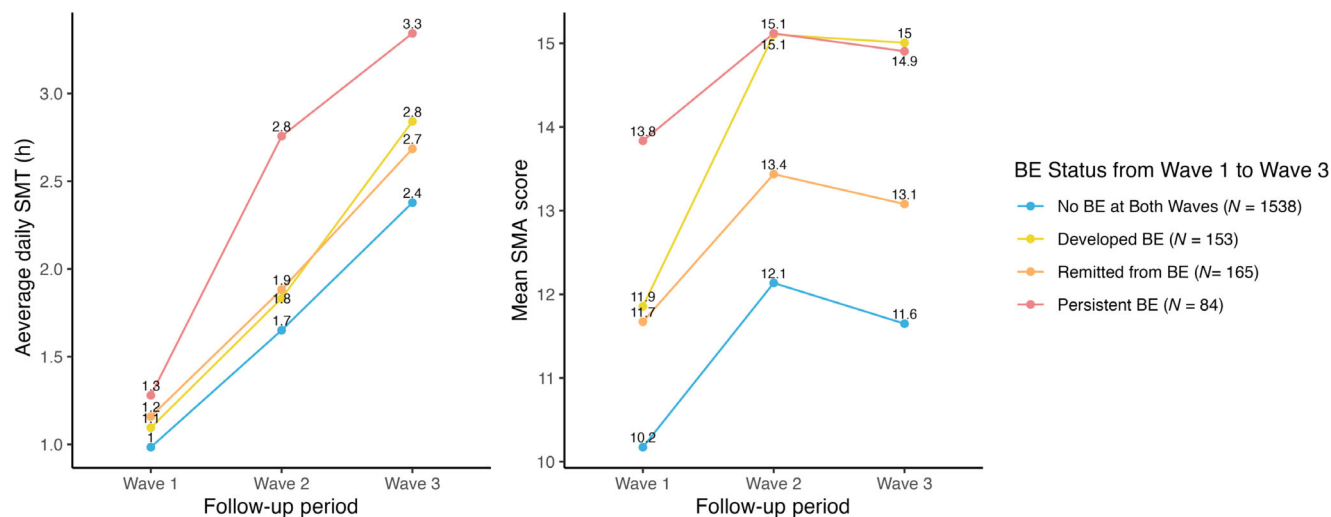
remitted from BE ( $n = 165$ ); and persistent BE ( $n = 84$ ). All BE profiles demonstrated increasing consumption of social media over the three waves, with daily SMT more than doubling by Wave 3 compared with Wave 1. Participants with persistent BE exhibited the largest increase, from 1.3 h/day at Wave 1 to 3.3 h/day at Wave 3. The SMA score peaked at the second wave for all four BE profiles. Participants with persistent BE maintained consistently high SMA scores, and participants who developed BE showed the most significant increase in SMA score.

Table 2 shows cross-sectional results. SMT was not cross-sectionally associated with BE at either Wave 1 or Wave 3. However, participants with mild SMA symptoms (vs. the asymptomatic group) had an increased prevalence of BE at Wave 3 (PR 1.73, 95% CI: 1.07–2.80), but not at Wave 1. Participants in the addiction SMA group showed increased BE prevalence at both waves, with PR of 2.17 (95% CI: 1.22–3.84) at Wave 1 and 3.45 (95% CI: 1.83–6.49) at Wave 3.

Longitudinal results (Table 3) found that SMT was not associated with Wave 3 BE. However, the addiction SMA group showed increased risk of BE at Wave 3 (RR 1.90, 95% CI: 1.10–3.27). No prospective associations were observed for the mild SMA symptom group.

Because a prospective association between SMA and BE was present, we proceeded to latent class trajectory modeling to identify SMA trajectory groups. The four-class model was selected based on its best balance between model fit and classification accuracy, as indicated by the lowest BIC and ICL values, as well as higher entropy (Figure S2). The following four groups emerged: consistently low SMA (reference group;  $n = 1263$ , 65.1%); decreased SMA ( $n = 455$ , 7.4%); increased SMA ( $n = 143$ , 23.5%); and consistently high SMA ( $n = 79$ , 4.1%; Figure 2).

The consistently high SMA group showed the strongest association with BE at Wave 3, with an RR of 3.01 (95% CI: 2.07–4.37) in the fully adjusted model. Both the decreased SMA and increased SMA groups showed comparable associations with follow-up BE, both before and after adjusting for the presence of BE at Wave 1. In the



**FIGURE 1** Trends in time spent on social media (SMT) and social media addiction (SMA) scores over three waves by four binge eating (BE) profiles. [Color figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

**TABLE 2** Cross-sectional associations of SMT and SMA with BE at Wave 1 or 3.

|               | Number of participants, n (%) | Crude model, PR (95% CI) | Model 1, PR (95% CI) | Model 2, PR (95% CI) |
|---------------|-------------------------------|--------------------------|----------------------|----------------------|
| Wave 1 SMT, h | 1940 (100)                    | 1.06 (1.00–1.12)         | 1.06 (1.00–1.12)     | 1.05 (0.99–1.12)     |
| Wave 3 SMT, h | 1940 (100)                    | 1.09 (1.05–1.14)         | 1.08 (1.03–1.13)     | 1.03 (0.98–1.09)     |
| Wave 1 SMA    |                               |                          |                      |                      |
| No symptoms   | 1408 (72.6)                   | Ref.                     | Ref.                 | Ref.                 |
| Mild symptoms | 477 (24.6)                    | 1.36 (1.01–1.84)         | 1.34 (0.99–1.82)     | 1.35 (1.00–1.84)     |
| Addiction     | 55 (2.8)                      | 2.40 (1.39–4.15)         | 2.25 (1.30–3.90)     | 2.17 (1.22–3.84)     |
| Wave 3 SMA    |                               |                          |                      |                      |
| No symptoms   | 1506 (77.6)                   | Ref.                     | Ref.                 | Ref.                 |
| Mild symptoms | 365 (18.8)                    | 1.84 (1.23–2.75)         | 1.75 (1.17–2.62)     | 1.73 (1.07–2.80)     |
| Addiction     | 69 (3.6)                      | 4.66 (2.79–7.77)         | 3.89 (2.32–6.51)     | 3.45 (1.83–6.49)     |

Note: Presence of BE symptoms was defined as any episode of losing control and consuming more than needed while eating in the past 2 weeks, as reported by either the child or parent. Based on the total score of the SMA questionnaire, participants were categorized into the following three groups: no symptoms (a score of 6, indicating no addictive symptoms); mild symptoms (scores of 7–23); and addiction (a score of 24 or higher, indicating problematic social media use). Model 1: Adjusted for sex, age, race and ethnicity, annual household income, and highest parental education; Model 2: Additionally adjusted for meeting behavioral recommendations (sleep duration, physical activity, and screen time) and BMI percentile, collected at the same wave as SMA/SMT and BE.

Abbreviations: BE, binge eating; PR, prevalence ratio; Ref., reference; SMA, social media addiction; SMT, time spent on social media.

fully adjusted model, the decreased SMA group had an RR of 1.92 (95% CI: 1.33–2.78), whereas the increased SMA group had an RR of 2.09 (95% CI: 1.59–2.74; Table 4). Among all participants with BE symptoms at Wave 3 ( $n = 237$ ), only those in the consistently high SMA group were more likely to be in the frequent BE symptom group at Wave 3 (RR 1.57, 95% CI: 1.04–2.38; Table S3).

### Stratification and sensitivity analyses

We found some evidence for an interaction with BMI percentile (Table S4) such that, among participants with overweight/obesity at Wave 1, those in the mild SMA symptom and addiction SMA groups at Wave 1, as well as those in the decreased SMA trajectory group, demonstrated a stronger association with BE at Wave 3 compared with their counterparts with underweight/healthy weight. Conversely, those in the consistently high SMA and increased SMA trajectory

groups showed a weaker association with BE at Wave 3 compared with the group with underweight/healthy weight (Table S5). Additionally, SMA showed a stronger prospective association with BE in male compared with female adolescents (Table S5).

Associations of SMT/SMA and BE based on child and parent reports were comparable with those for combined reports except for the increased SMA group (Table S6). Additionally, consistent with the main analysis, neither cross-sectional nor longitudinal associations of SMT and BE were significant in sensitivity analyses including participants who did not report having their own social media account (Table S7).

### DISCUSSION

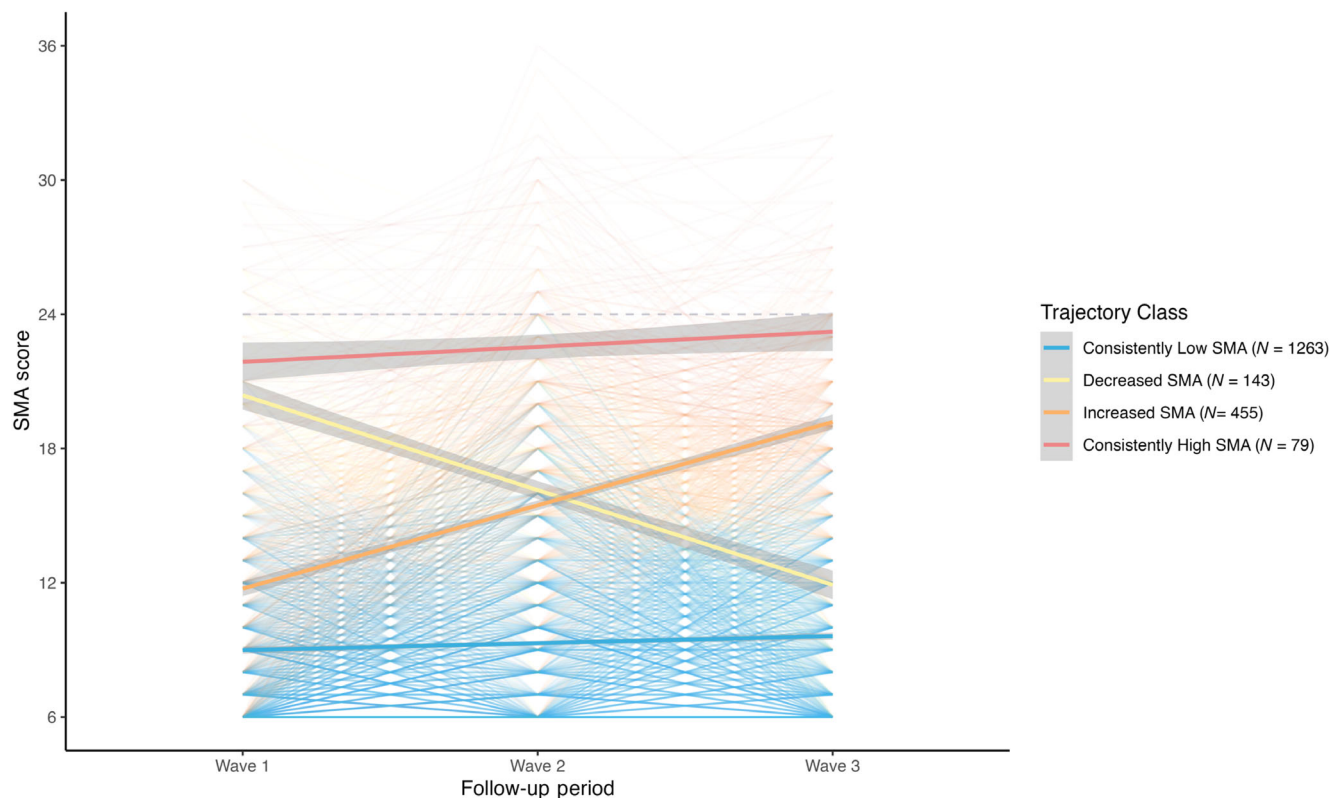
Using data from the ABCD Study, a large early adolescent cohort of US children, we analyzed concurrent and longitudinal relationships of

**TABLE 3** Prospective associations of Wave 1 SMT and SMA with Wave 3 BE.

|               | Crude model, RR (95% CI) | Model 1, RR (95% CI) | Model 2, RR (95% CI) | Model 3, RR (95% CI) |
|---------------|--------------------------|----------------------|----------------------|----------------------|
| Wave 1 SMT, h | 1.05 (0.99–1.10)         | 1.03 (0.97–1.09)     | 1.01 (0.94–1.08)     | 0.99 (0.93–1.06)     |
| Wave 1 SMA    |                          |                      |                      |                      |
| No symptoms   | Ref.                     | Ref.                 | Ref.                 | Ref.                 |
| Mild symptoms | 1.56 (1.13–2.16)         | 1.57 (1.14–2.17)     | 1.46 (1.06–2.00)     | 1.37 (1.00–1.87)     |
| Addiction     | 2.82 (1.61–4.93)         | 2.64 (1.51–4.62)     | 2.23 (1.27–3.90)     | 1.90 (1.10–3.27)     |

Note: Presence of BE symptoms was defined as any episode of losing control and consuming more than needed while eating in the past 2 weeks, as reported by either the child or parent. Based on the total score of the SMA questionnaire, participants were categorized into the following three groups: no symptoms (a score of 6, indicating no addictive symptoms); mild symptoms (scores of 7–23); and addiction (a score of 24 or higher, indicating problematic social media use). Model 1: Adjusted for sex, age, race and ethnicity, annual household income, and highest parental education; Model 2: Additionally adjusted for meeting behavioral recommendations (sleep duration, physical activity, and screen time) and BMI percentile, collected at the same wave as SMA/SMT and BE; Model 3: Additionally adjusted for BE status at Wave 1.

Abbreviations: BE, binge eating; RR, relative risk; SMA, social media addiction; SMT, time spent on social media.



**FIGURE 2** Trajectories of social media addiction (SMA) scores over Waves 1 to 3. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/oby.24199)]

two different measures of social media usage with BE. Our findings revealed that SMA scores, but not SMT, were prospectively associated with BE, with the strongest effects for individuals with consistently high SMA and similar but smaller effects for individuals with either increasing or decreasing SMA, suggesting that a history of addictive symptoms in relation to social media is linked to BE at follow-up regardless of current social media-addictive behaviors. This association was particularly evident in male adolescents and varied by baseline BMI percentile.

The concurrent association of high SMA scores and BE was more prominent at Wave 3, when 13- to 14-year-old participants were

entering adolescence and beginning to undergo puberty-associated changes that increased psychological and behavioral vulnerability. Several factors may contribute to the peaks of SMA and BE and the association of SMA and BE that we observed at this age. First, weight concerns and self-body image dissatisfaction develop with and increase through puberty [28]. Heightened, persistent distress toward one's own body weight/shape could both predispose to BE onset (or exacerbate it if already present) [29] and facilitate the overuse of social media [30]. Social media, in turn, exhibiting idealized lives and bodies, may have a detrimental effect on well-being [31] by increasing comparisons, appearance-based evaluations, and body dissatisfaction.

**TABLE 4** Associations between SMA score trajectories over three waves and Wave 3 BE.

|               | Crude model, RR (95% CI) | Model 1, RR (95% CI) | Model 2, RR (95% CI) | Model 3, RR (95% CI) |
|---------------|--------------------------|----------------------|----------------------|----------------------|
| Low SMA       | Ref.                     | Ref.                 | Ref.                 | Ref.                 |
| Decreased SMA | 2.52 (1.72–3.70)         | 2.50 (1.70–3.68)     | 2.28 (1.56–3.33)     | 1.92 (1.33–2.78)     |
| Increased SMA | 2.46 (1.88–3.22)         | 2.42 (1.84–3.17)     | 2.31 (1.77–3.03)     | 2.09 (1.59–2.74)     |
| High SMA      | 3.92 (2.67–5.75)         | 3.65 (2.47–5.39)     | 3.36 (2.28–4.95)     | 3.01 (2.07–4.37)     |

**Note:** Presence of BE symptoms was defined as any episode of losing control and consuming more than needed while eating in the past 2 weeks, as reported by either the child or parent. Based on total scores on the SMA questionnaire, participants were categorized into the following three groups: no symptoms (a score of 6, indicating no addictive symptoms); mild symptoms (scores of 7–23); and addiction (a score of 24 or higher, indicating problematic social media use). Model 1: Adjusted for sex, age, race and ethnicity, annual household income, and highest parental education; Model 2: Additionally adjusted for meeting behavioral recommendations (sleep duration, physical activity, and screen time) and BMI percentile, collected at the same wave as SMA/time on social media and BE; Model 3: Additionally adjusted for BE status at Wave 1.

Abbreviations: BE, binge eating; RR, relative risk; SMA, social media addiction.

However, we note that positive effects such as offering inspiring models that facilitate solidarity [32] and providing emotional regulation [33] also occur. We additionally note that neural circuits undergirding reward responsivity and cognitive control undergo significant development from early through later adolescence [34], making early adolescence a vulnerable period for developing and maintaining dysfunctional behaviors to rewarding stimuli such as food or social media [35].

Contrary to a previous study of the ABCD cohort that found a prospective association between SMT and BED [10], our study did not find a link between SMT and BE symptoms. Several factors may explain this discrepancy. The prior study relied on eating disorder diagnoses, which were subsequently identified as inaccurate in the ABCD Study 5.1 release note [36], whereas we focused on symptom-based variables. Due to confining our analysis to cases with complete data, our study used an analysis subsample of the ABCD cohort, which differed in baseline age, sex distribution, socioeconomic status, and cultural background from the broader sample, potentially influencing our results. Additionally, the previous study reported that the effects of SMT on BED were significant but small, suggesting that their findings might be sensitive to sample characteristics and analysis methods. Instead, we observed associations between BE and pathological aspects of social media use, specifically SMA, which could potentially contribute as emotional dysregulation and impulsivity, key factors in disordered eating [37, 38]. We speculate that SMA might be more relevant than SMT in relation to BE. However, the relatively low prevalence of SMA in our sample at both waves is a limitation and highlights the need for cautious interpretation of these findings.

Our analyses of moderation of the prospective association between BE and SMA identified specific at-risk populations. First, male adolescents showed a greater association between SMA and BE than female adolescents. These findings align with existing evidence suggesting that male adolescents are more susceptible to addictive behaviors due to higher levels of impulsivity exhibited and the tendency to externalize behaviors [39]. Furthermore, male adolescents have been shown to rarely seek social support and to engage more readily in dysfunctional, avoidant coping strategies to face stressors [40]. Second, in partial support of our hypothesis, BMI percentile interacted with SMA to influence BE, but this occurred in a complex manner, with analyses stratifying by weight group demonstrating that participants who had overweight/obesity compared with underweight/healthy weight at baseline showed only inconsistent evidence for a stronger association between SMA and BE. We note that 15.8% and 8.9% of participants in these groups, respectively, shifted between underweight/healthy weight and overweight/obesity categories during the study period, and it is possible that BMI trajectories rather than baseline BMI may be the more meaningful moderator of relationships between SMA and BE. In this study, we did not include weight changes in the model because it was not our primary focus, opting instead for conventional control of confounders by baseline adjustment [41]. However, future research should consider the time-varying effects of weight and implement nonlinear models to gain a fuller understanding of how baseline weight and/or weight changes

might influence the relationships between SMA and BE. We note that, as the ABCD cohort ages, social media use, BE, and rates of overweight/obesity will likely increase, facilitating further research on interrelationships into later adolescence.

Both decreased and increased SMA groups showed comparable associations with follow-up BE, suggesting that an initial history of addictive behaviors, regardless of its subsequent trajectory, may be linked to a higher prevalence of BE later on. This finding aligns with research showing that early experience with addictive behaviors can have long-term impacts on behavior due to changes in brain plasticity caused by repeated stimulation of the mesolimbic dopamine system [42] and suggests that potentially protective effects of early limits on social media exposure on later development of disordered eating should be explored. We also emphasize that the relationships that we observed may not demonstrate a causal effect of social media-addictive behaviors on BE but instead demonstrate a process by which a shared biobehavioral vulnerability manifests in a co-occurrence of both behaviors. Our results support further investigation not only of potential causal relationships between social media-addictive behaviors and BE but also efforts to identify and intervene on risk factors and influences on multiple co-occurring addictive behaviors.

The major strength of our study is the relatively large sample size, enabling a robust analysis of the prospective association between social media use and BE while distinctly evaluating the impacts of SMT and SMA. However, we acknowledge several limitations. First, the self-reports of SMT that were used are subject to recall bias and social desirability bias and did not allow exploration of the effects of social media content. Second, the regulatory US minimum age requirement of 13 years for creating social media accounts falls within the early adolescent stage. Because our main analysis was restricted to participants who reported having accounts at all three waves to minimize bias introduced by having a social media account at one wave but not another, interpretations concerning age should be approached cautiously because those who reported circumventing this age threshold might display unmeasured characteristics that could skew the findings. However, we did observe similar findings for SMT in a sensitivity analysis of a larger sample that included participants who did not formally report having a social media account. Third, although we used a combined report from both parents and children, the ABCD protocol does not constitute a gold-standard clinical assessment of BE, and the symptoms recorded do not necessarily indicate a formal diagnosis of BED. Fourth, although we found a prospective association, this does not entail causation, and the observed relationships may be influenced by unmeasured confounding variables and fail to fully capture the time course of longitudinal relationships due to the sparse measurement schedule. Fifth, generalizability of our findings may be limited due to the specific demographic and geographic characteristics of the study sample, as well as the timing of the data collection, some of which occurred during and subsequent to the COVID-19 pandemic. Finally, we emphasize that SMA is not currently recognized as a formal diagnosis in the DSM-5-TR and is

instead defined in the current study using a self-report research instrument [21], highlighting the need for cautious interpretation of our results.

## CONCLUSION

Limitations notwithstanding, our findings suggest that social media-addictive behaviors may serve as an early indicator of subsequent BE. This supports further research to explore whether optimizing social media usage and promoting healthier online behaviors in early adolescence could help reduce the burden of BE. Managing addictive behaviors rather than focusing solely on SMT may be particularly beneficial, especially for male adolescents, and effects may also differ by body weight. Further observational and interventional studies are needed to better understand the underlying mechanisms and determine whether SMA could predict broader disordered behaviors, as well as whether managing social media-addictive behaviors could help mitigate the risk of disordered eating and other problematic behaviors. 

## AUTHOR CONTRIBUTIONS

Xinyi Shi: writing of original draft; writing, review and editing; visualization; formal analysis; data curation; and conceptualization. Sarah Ann Duck: writing, review and editing, and data curation. Elena Jansen: writing, review and editing. Bianca Borsarini: writing, review and editing. Courtney K. Blackwell: writing, review and editing. Yuchen Li: writing, review and editing. Susan Carnell: writing, review and editing; supervision; funding acquisition; and conceptualization.

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## CONFLICT OF INTEREST STATEMENT

The authors declared no conflicts of interest.

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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